Architecture and Cost-Benefit Analysis of a High-Precision Synchronized SDR System

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February 3, 2026

Abstract

This document outlines the technical specifications, Bill of Materials (BOM), and design trade-offs for a software-defined radio (SDR) system optimized for phase-coherent applications, such as direction finding and precision signal acquisition. The architecture integrates an Analog Devices AD936x-based transceiver with a Jetson Nano for edge processing, utilizing a GPS-Disciplined Oscillator (GPSDO) to ensure frequency stability and temporal alignment via 1 PPS and 10 MHz reference signals.

1 System Architecture

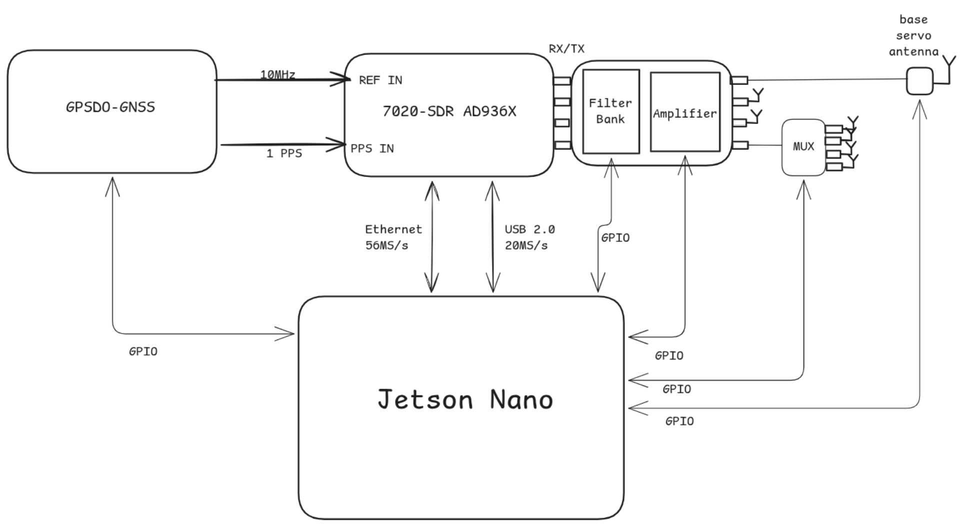

Figure 1: Functional Block Diagram: Synchronized SDR with RF Front-End and Edge Compute.

2 Bill of Materials (BOM)

The following table provides the components required for two distinct configurations. All prices are in USD and were verified on February 3, 2026.

Component	Technical Justification	Price (USD)	Ref.
SDR (Entry-Level)	Nooelec NESDR Smart v5: Reliable wideband receiver (100kHz–1.75GHz).	$42.95	[1]
SDR (Research-Grade)	OpenSourceSDRLab 7020-SDR AD9361: Zynq 7020 based platform with MIMO.	$498.00	[2]
LNA (Front-end)	Nooelec LaNA: Wideband Low-Noise Amplifier (20MHz–4GHz).	$29.95	[3]
GPSDO Reference	Leo Bodnar Mini GPSDO: Ultra-stable 10 MHz reference.	$125.00	[4]
Antenna (General)	Wideband Log-Periodic (600 MHz–6 GHz) for survey work.	$45.00	[5]
Servo Antenna Base	Pan-Tilt Servo mechanism for directional tracking.	$150.00	[6]
Support Hardware	SMA Cables, 5V/4A Power Supply, and GPIO Breakouts.	$80.00	[7]

2.1 Cost Summation

- **Entry-Level Configuration Total:** $472.90
- **Research-Grade Configuration Total:** $927.95

3 Design Purpose

This system is designed for **Radio Frequency Interference (RFI) Localization** and **Time Difference of Arrival (TDOA)** research. By utilizing the 1 PPS signal from the GPSDO, the system can timestamp IQ samples with nanosecond precision.

4 Weaknesses and Limitations

1. **Dynamic Range:** The AD9361 uses a 12-bit ADC.
2. **Host Compute Dependency:** Jetson Nano throughput limitations over USB 2.0.
3. **Calibration:** Phase coherence requires periodic thermal calibration.
4. **Susceptibility:** Servo motor EMI requires shielding.

5 Assumptions and Verifications

- **Prices:** Market listings as of February 2026.
- **Specifications:** Bandwidth limits are based on hardware bus capabilities.

[1] Amazon, Feb 3, 2026. https://www.amazon.com/Nooelec-RTL-SDR-SDR-100kHz-1-75GHz-Enclosure/dp/B01HA642SW

[2] OpenSourceSDRLab, Feb 3, 2026. https://opensourcesdrlab.com/products/opensourcesdrlab-new-7020-sdr-ad9361-development-board-with-pa-for-pluto-sdr-matlab-software-defined-radio?VariantsId=10346

[3] Nooelec, Feb 3, 2026. https://www.nooelec.com/store/lana.html

[4] Leo Bodnar, Feb 3, 2026. https://www.leobodnar.com/shop/index.php?main_page=product_info&cPath=107&products_id=393

[5] Generic RF supplier average price, Feb 2026.

[6] Estimated price for precision metal-gear kit, Feb 2026.

[7] Estimated based on current hardware listings.